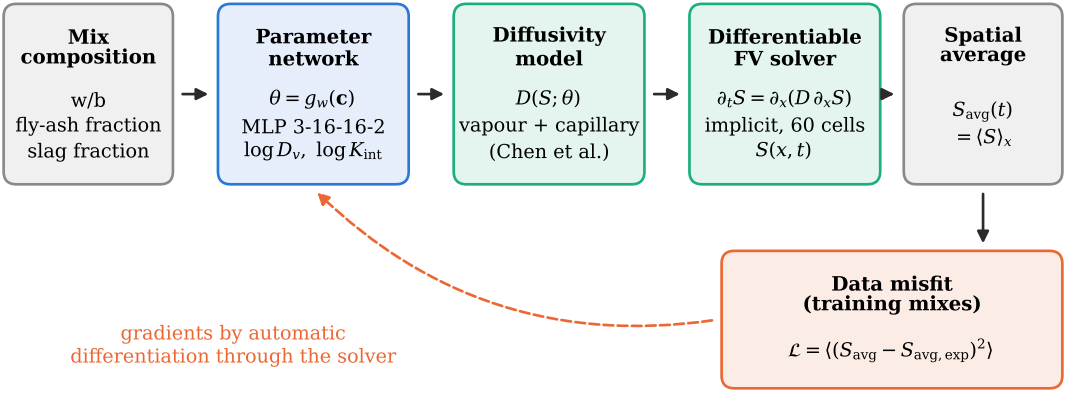
Mix composition
w/b
fly-ash fraction
slag fraction
Parameter network
$\theta = g_w(\mathbf{c})$
MLP 3-16-16-2
$\log D_v,\ \log K_{\text{int}}$
Diffusivity model
$D(S;\theta)$
vapour + capillary
(Chen et al.)
Differentiable FV solver
$\partial_t S = \partial_x(D\,\partial_x S)$
implicit, 60 cells
$S(x,t)$
Spatial average
$S_{\text{avg}}(t)$
$= \langle S \rangle_x$
Data misfit (training mixes)
$\mathcal{L} = \langle (S_{\text{avg}} - S_{\text{avg,exp}})^2 \rangle$
gradients by automatic
differentiation through the solver